

Computer Organization And Assembly Language

```
00000000: 01001101 01011010 10010000 00000000 00000011 00000000 MZ....
00000006: 00000000 00000000 00000100 00000000 00000000 00000000 .....
0000000c: 11111111 11111111 00000000 00000000 10111000 00000000 .....
00000012: 00000000 00000000 00000000 00000000 00000000 00000000 .....
00000018: 01000000 00000000 00000000 00000000 00000000 00000000 @....
0000001e: 00000000 00000000 00000000 00000000 00000000 00000000 .....
00000024: 00000000 00000000 00000000 00000000 00000000 00000000 .....
0000002a: 00000000 00000000 00000000 00000000 00000000 00000000 .....
00000030: 00000000 00000000 00000000 00000000 00000000 00000000 .....
00000036: 00000000 00000000 00000000 00000000 00000000 00000000 .....
0000003c: 10000000 00000000 00000000 00000000 00001110 00011111 .....
00000042: 10111010 00001110 00000000 10110100 00001001 11001101 .....
00000048: 00100001 10111000 00000001 01001100 11001101 00100001 !..L.!
0000004e: 01010100 01101000 01101001 01110011 00100000 01110000 This p
00000054: 01110010 01101111 01100111 01110010 01100001 01101101 rogram
0000005a: 00100000 01100011 01100001 01101110 01101110 01101111 canno
00000060: 01110100 00100000 01100010 01100101 00100000 01110010 t be r
00000066: 01110101 01101110 00100000 01101001 01101110 00100000 un in
0000006c: 01000100 01001111 01010011 00100000 01101101 01101111 DOS mo
00000072: 01100100 01100101 00101110 00001101 00001101 00001010 de...
00000078: 00100100 00000000 00000000 00000000 00000000 00000000 $.
0000007e: 00000000 00000000 01010000 01000101 00000000 00000000 ..PE..
```

Lab Manual 11

Objectives:

1. Revision of mul and div instructions.
2. Input and output using mul and div instructions
3. Understanding Program Segment Prefix
4. Understanding and working with command line arguments.

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Command Line Arguments

Command line arguments are values passed to a program when it is executed in a command-line environment. They provide a way for users to influence the behavior of a program by specifying additional information at runtime. In assembly language programming, command line arguments are often accessed through registers or memory locations, depending on the specific architecture. Typically, command line arguments are passed to a program as strings. These strings can represent various types of data, such as integers, floating-point numbers, or even file names. The program needs to parse and interpret these strings to extract meaningful information.

Task 1

Write an assembly program that takes two integer parameters from the command line and prints their sum in binary, decimal and hexadecimal.

Task 2

Develop an assembly program that accepts three command line arguments: two integers and an arithmetic operation (addition, subtraction). Perform the specified operation on the integers and display the result in binary, decimal and hexadecimal.

Note: Write only one procedure for displaying output and call it multiple times for output

Task 3

Write an assembly language program that will perform like dump instruction in the debugger. Use different versions of this command including command line arguments to handle starting and ending offset addressed to display the data.

Some examples of addresses for dump instruction may include.

- 201:200
- 0231:0200
- 230 240
- 230:200 230

You can refer to lab manual 1 for further details of dump instruction, and use code developed for displaying data from previous labs as well.